Saintgits College of Engineering

Department of Electronics and Communication Engineering

DIGITAL SYSTEMS LAB · PROJECT REPORT

Digital Railway Track Collision Prevention System

Using Logic Gates

A microcontroller-free demonstration of conflict detection using AND and NOT gates, LEDs and a buzzer

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1. Abstract

This project shows how basic logic gates can detect a conflict when two trains enter the same railway crossing. Two push buttons stand in for Train A and Train B. A 74HC08 AND gate checks whether both are active at the same time. If they are, a red LED turns on and a buzzer sounds to signal a potential collision. A 74HC04 NOT gate inverts the same signal to drive a green LED, which shows that no conflict is detected.

The circuit is built on a breadboard from digital ICs, resistors, a transistor and a regulated 5 V supply, without an Arduino or any other microcontroller. It is a simplified educational model of conflict detection, not a certified railway safety system.

2. Introduction

When two trains enter a shared section of track at the same time, a collision can occur. Railway signalling therefore has to recognise when two trains are about to occupy the same section. This project demonstrates that idea at the lowest level of digital electronics, using logic gates instead of a processor.

Each push button acts as a sensor that reports a train entering the shared area. The logic gates compare the two reports and decide whether they overlap. The result is shown on two LEDs and a buzzer. Using gates instead of a microcontroller keeps the behaviour fully visible: every output follows from the inputs through a simple Boolean function that can be checked with a truth table.

3. Objectives

- To demonstrate a practical application of digital logic gates.
- To detect a conflict when both train inputs are active.
- To indicate the safe and conflict conditions using LEDs and a buzzer.
- To understand how digital circuits can be applied to railway signalling concepts.

4. Components Required

The circuit uses the components below. Both gate ICs are from the 74HC (CMOS) family, so a 10 kΩ pull-down resistor on each input gives a clean LOW level.

Sl. No.	Component	Specification	Qty	Purpose
1	AND gate IC	74HC08	1	Detects both trains present
2	NOT gate IC	74HC04	1	Drives the green LED
3	Push buttons	Normally open	2	Simulate Train A and Train B
4	Red LED	5 mm	1	Conflict indicator
5	Green LED	5 mm	1	No-conflict indicator
6	Active buzzer	5 V, low current	1	Audible warning
7	NPN transistor	BC547	1	Switches the buzzer
8	Resistor	10 kΩ	2	Pull-down for button inputs
9	Resistor	1 kΩ	2	LED current limiting
10	Resistor	4.7 kΩ	1	Transistor base current limiting
11	Ceramic capacitor	100 nF	2	Supply decoupling, one per IC
12	Breadboard	Solderless	1	Circuit assembly
13	Jumper wires	Male-to-male	10–20	Connections
14	Power supply	Regulated 5 V DC	1	Power for the circuit
15	Cardboard or foam board	For the model	1	Railway model

Table 1: Bill of materials

5. Working Principle

The red LED and the buzzer turn on only when both train inputs are HIGH. In every other case the green LED is on. Let A be the Train A input, B the Train B input, C the conflict signal and G the green (no-conflict) signal. The circuit implements two Boolean equations:

$$C = A \cdot B$$

$$G = \overline{A \cdot B}$$

Train A	Train B	Green LED	Red LED	Buzzer
OFF (0)	OFF (0)	ON	OFF	OFF
ON (1)	OFF (0)	ON	OFF	OFF
OFF (0)	ON (1)	ON	OFF	OFF
ON (1)	ON (1)	OFF	ON	ON

Table 2: Truth table of the system

How each case behaves. With no train present, both inputs are LOW, the AND output is LOW and the NOT gate output is HIGH, so only the green LED lights. With one train present, the other input is still LOW, so the result is the same. When both buttons are pressed, the AND output goes HIGH: the red LED turns on, the transistor switches the buzzer on, and the NOT gate output goes LOW, which turns the green LED off. Releasing either button clears the conflict.

The green LED means only that no conflict is detected. It does not prove that the track is safe.

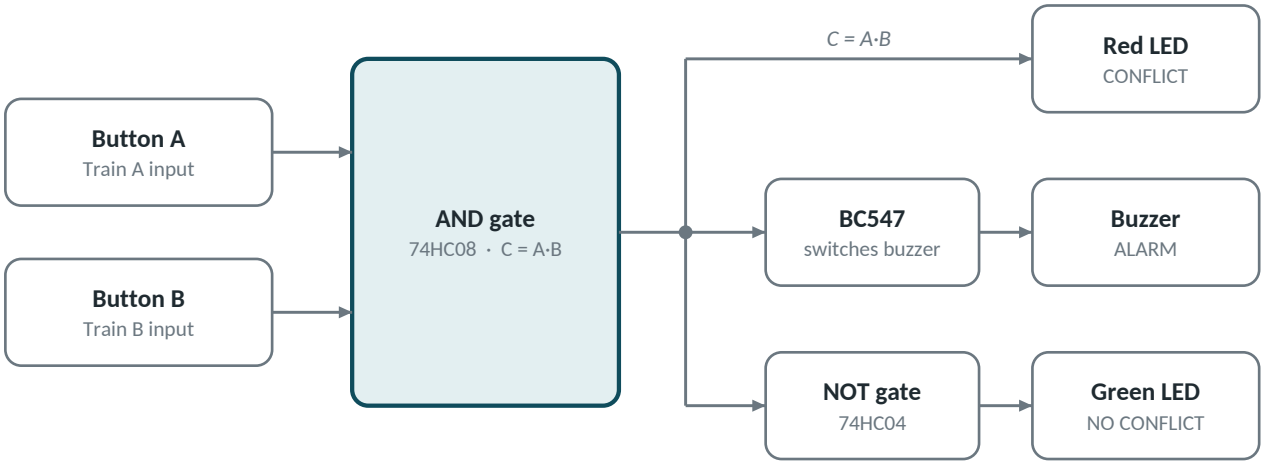


Figure 1: Functional block diagram of the system

6. Circuit Description

The circuit has three sections.

- **Input section:** two normally open push buttons connect +5 V to the gate inputs. A 10 kΩ pull-down resistor on each input holds it LOW while the button is released, so the input is never left floating. A pressed button gives HIGH.
- **Logic section:** the first gate of the 74HC08 (pins 1 and 2 in, pin 3 out) forms C = A·B. The first inverter of the 74HC04 (pin 1 in, pin 2 out) forms G, the inverse of C.
- **Output section:** the red LED is driven directly from the AND output through a 1 kΩ resistor, and the green LED from the NOT output through another 1 kΩ resistor. A BC547 transistor lets the AND output switch the buzzer without the IC supplying its full current.

Role of the supporting components

Component	Value	Function
Pull-down resistors	10 kΩ × 2	Hold each button input at a defined LOW when released
LED resistors	1 kΩ × 2	Limit LED current to about 3 mA (assuming a 2 V LED drop)
Base resistor	4.7 kΩ	Limits BC547 base current to about 0.9 mA (assuming $V_{BE} \approx 0.7$ V)
BC547 transistor	NPN	Acts as a switch so the buzzer current does not flow through the gate
Decoupling capacitors	100 nF × 2	One across pins 14 and 7 of each IC to reduce supply disturbances

Table 3: Supporting components

7. Circuit Diagram and Connections

Figure 2 shows the complete schematic. Each IC is powered from pin 14 (+5 V) and pin 7 (GND), with a 100 nF capacitor across those two pins placed close to the IC. Figure 3 shows the pin assignments of both ICs.

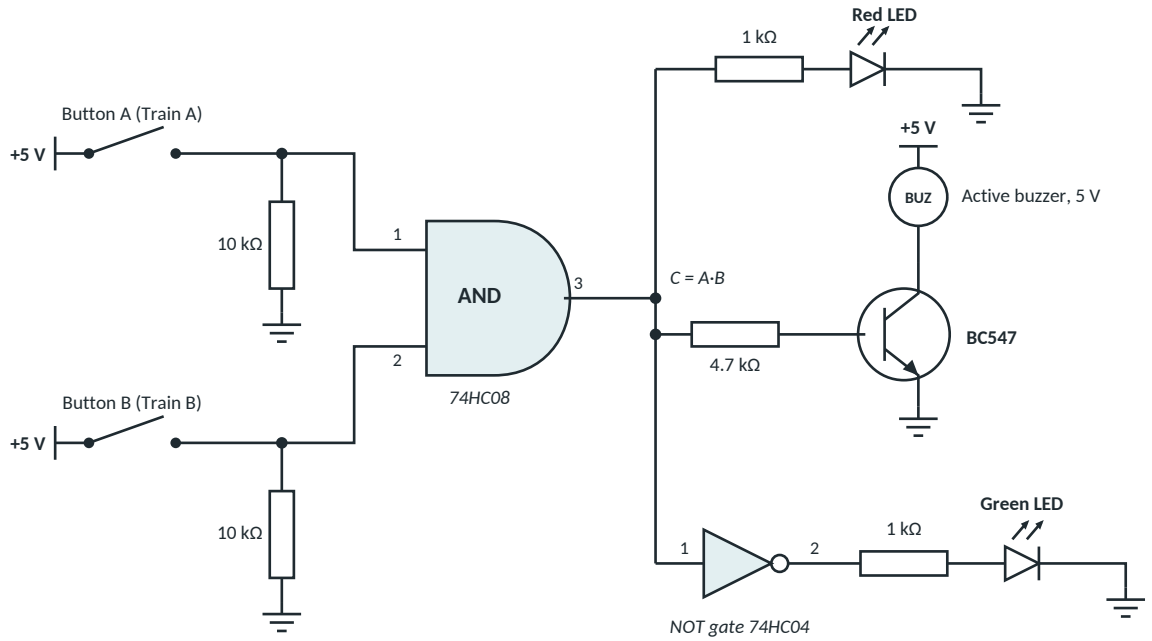


Figure 2: Circuit diagram (power pins and decoupling capacitors not shown; see Figure 3 and Table 5)

Wiring checklist

From	Through	To
+5 V	Button A	74HC08 pin 1
74HC08 pin 1	10 kΩ	GND
+5 V	Button B	74HC08 pin 2
74HC08 pin 2	10 kΩ	GND
74HC08 pin 3	1 kΩ	Red LED anode (cathode to GND)
74HC08 pin 3	Direct wire	74HC04 pin 1
74HC04 pin 2	1 kΩ	Green LED anode (cathode to GND)
74HC08 pin 3	4.7 kΩ	BC547 base
BC547 collector	Direct wire	Buzzer negative terminal
Buzzer positive terminal	Direct wire	+5 V
BC547 emitter	Direct wire	GND
Pin 14 of each IC	100 nF	Pin 7 of the same IC

Table 4: Wiring checklist

IC pin connections

CMOS inputs must never be left floating, so every unused gate input is tied to GND. Unused outputs are left open and are never connected to GND.

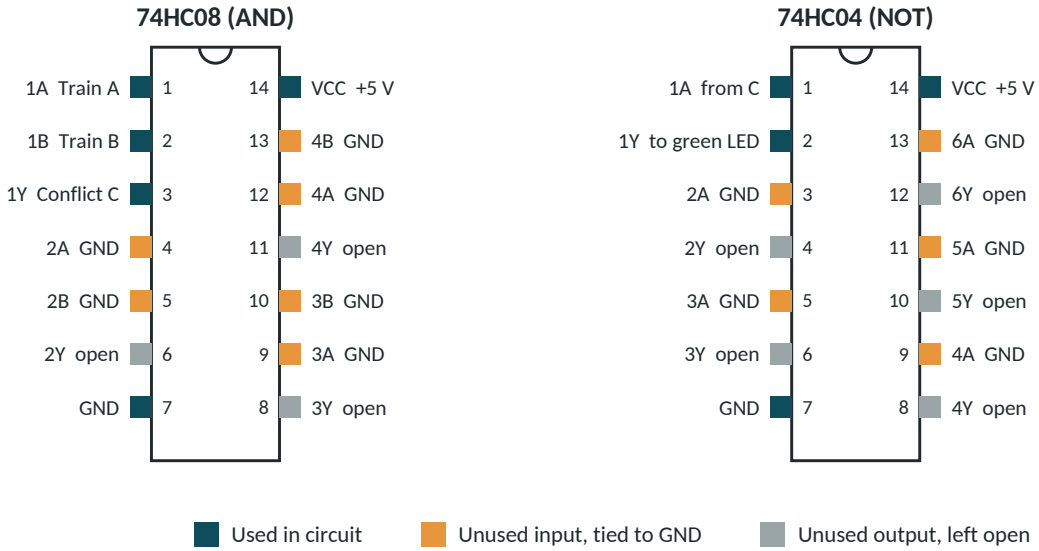


Figure 3: Pinouts of the 74HC08 and 74HC04 (top view, notch at top)

IC	Unused inputs (tie to GND)	Unused outputs (leave open)
74HC08	Pins 4, 5, 9, 10, 12, 13	Pins 6, 8, 11
74HC04	Pins 3, 5, 9, 11, 13	Pins 4, 6, 8, 10, 12

Table 5: Unused pins

Check before powering on: verify the pinout of your own BC547 (the order depends on the package and manufacturer), confirm the buzzer is an active 5 V type within the transistor's current limits, check each IC's orientation using its notch, and never connect +5 V directly to GND.

8. Testing Procedure and Expected Results

The electronics are tested on the breadboard before the railway model is built. Each of the four input combinations is applied in turn and the outputs are compared with the truth table.

Test	Action	A	B	Expected result
1	Release both buttons	0	0	Green LED on; red LED and buzzer off
2	Press button A	1	0	Green LED on; red LED and buzzer off
3	Release A, press button B	0	1	Green LED on; red LED and buzzer off
4	Hold A, then press B too	1	1	Green LED off; red LED on; buzzer sounds

Table 6: Test cases

If all four tests give the expected result, the circuit is operating as designed. Releasing either button in test 4 clears the conflict indication.

9. Railway Model

To make the demonstration easier to follow, the tested circuit is mounted on a cardboard railway model. The physical trains need no motors, because the buttons represent the sensors.

- Two miniature tracks made with black lines or strips.
- Two small paper trains labelled Train A and Train B.
- One push button beside each train, labelled Train A Sensor and Train B Sensor.
- Green and red LEDs labelled NO CONFLICT DETECTED and CONFLICT.
- A small warning sign beside the red LED.

10. Applications

The same AND-gate principle applies wherever two conditions must not occur together.

- Digital railway signalling concepts and conflict detection.
- Monitoring systems and warning circuits.

- Safety interlocks.

11. Limitations

This is a digital conflict-detection demonstration and not a complete railway collision-prevention system.

- Push buttons stand in for sensors, and each button is momentary.
- The circuit only warns. It does not stop or control any train.
- It has no fail-safe behaviour: if power or an input fails, no warning appears.
- It assumes the two inputs represent trains entering the same conflicting area. Real railway systems need additional sensing, interlocking and certified fail-safe design.

12. Future Scope

- Latch the conflict indication with a flip-flop until a reset button is pressed.
- Replace the buttons with IR or reed-switch sensors on a model track.
- Add more track sections using extra gates.
- Drive a relay or servo to operate a crossing barrier.

13. Conclusion

The project shows that an AND gate and a NOT gate are enough to detect a specific condition and report it with both visual and audible warnings. It also covered the practical details of working with CMOS logic: pull-down resistors, tying unused inputs to ground, supply decoupling, and using a transistor to drive a load that a gate cannot. The circuit detects a conflict but does not prevent one, so it remains a simplified educational model of how digital logic supports railway signalling.

14. References

1. Digital Railway Track Collision Prevention System, Digital Systems Lab Project Guide.
2. M. Morris Mano and Michael D. Ciletti, *Digital Design*, Pearson.
3. *74HC08; 74HCT08 Quad 2-input AND gate*, manufacturer datasheet.
4. *74HC04; 74HCT04 Hex inverter*, manufacturer datasheet.
5. *BC547 NPN general-purpose transistor*, manufacturer datasheet.